

Live Detection of Foreign Object Debris on Runways

Detection using Drones and AI

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Abstract – The removal of foreign objects and debris (FOD) on runways at both non-towered and tower-controlled airstrips is a universal problem. Current detection systems are relatively efficient at finding large foreign objects but are expensive to set up and tend to be inaccurate with small items. With recent developments in artificial intelligence and object detection, it has been proposed to use drones to collect images and then detect any FODs with a specific trained algorithm. This paper will discuss the required materials and method, the image collection process as well as the training and testing results analysis. The development and testing of a live detection model is also discussed. Six object tags have been selected for testing that are realistic representations of possible objects found on runways, these include *Litter, Wood, Bolts, Metal, Rubber* and *Nuts*. The collected image database provided a larger variety of images to increase the robustness of the model.

The Microsoft Azure platform and Custom Vision module was selected for the deep learning and object detection component. Through the system development, two different models were created, the first with six individual classes and then a single object class model. Two different training algorithms were also used, the online general model and the local compact model. The *compact* model is able to rapidly analyse images however it loses overall model accuracy. The *general* model is significantly more accurate than the compact model, however it requires significantly more time to analyse. The *single class general* trained model has the lowest error rate of 10.5% with 51 correct objects detected out of 57.

A live detection module was developed using a 4K HDMI video capture card to live stream the drone footage from the controller to the computer. The individual frames were then extracted and loaded through the detection module. The returned objects that have a probability above the set threshold are processed with a bounding box and given classification tag names. A GPS transmitter and receiver model captured the current coordinates of the drone and then synchronises it with the live video frame data. This was used to create a user-friendly data representation that is accurate to approximately 2.5m.

Keywords — *Runway Foreign Object and Debris Detection, Drone Imaging of Runways, Live Detection with GPS Module, Custom Vision (Microsoft Azure)*

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INTRODUCTION

The danger of foreign objects causing damage on aeroplanes during take-off and landing is considered very serious. Foreign object (FOD) debris is considered any object found in an inappropriate/undesired location, that have the potential to cause damage to personnel or equipment. The range of possible objects include natural elements such as rocks, wildlife and loose pavement, plus man-made objects like pieces of luggage, hardware, tools or broken equipment. [1]

The effects and damage caused by foreign objects is estimated to cost the aerospace industry \$3-4 billion dollars a year in repairs and maintenance. This value does not take into account the cost or loss of revenue caused by flight delays and schedule changes. The loss of engine efficiency during flight caused by damage from FODs can cost up to three times the normal running cost. [2]

In this paper we focused on the detection of manmade objects and materials in airport scenarios instead of the possibility of wildlife located around an airport which can be just as detrimental to aircraft. The current common detection method requires airport staff to manually search and remove objects. At the Brisbane International and Domestic Airport (QLD, Australia) there are approximately 30 people working around the clock to maintain and inspect the taxiways and runways. [3]. A

wide range of manual tools are used to assist in the detection and collection of FOD; however, they are all limited to the accuracy of the human user and time available.

There are currently several patented mechanical methods to detect FOD located on runways, some examples include radar, infrared technologies, and stationary RGB cameras. These are mounted along the runway and use image processing tools to detect objects to then warn the operators. For a detection system to be viable, it is required to be accurate in all weather conditions and able to detect a wide range of possible objects. It must also be able to examine the complete length of the runway and taxiways during operation. [4], [5], [6]

The radar technology is highly accurate when finding large FODs but is highly inaccurate with small items causing a high false-positive rate. Stationary based RGB camera-based methods also have a high false-positive rate prompting the shutdown of runways, this creates disruptions for both the airport and the airline carriers. The aerodynamic design and application of drones and AI on the other hand has seen a range of applications. [7], [8], [9]

The concept presented in this paper uses a drone to capture the images with deep learning and AI program to detect foreign objects at airports including at the terminal, taxiway or runways. The deep learning object identification algorithm is developed using the Microsoft Azure Custom Vision module. The accuracy and robustness of the model is paramount for the final product. The image resolution and effective drone height are critical components. To minimise the time required to analyse the runway, taxiways and other traffic paths within an airport, a live detection process was created and tested.

The high-level concept of operations for the system is for the drone to fly from one end of the runway to the other multiple times as required to capture images of the surface. Throughout the flight the drone will be capturing high resolution video and sending the feed directly back to the user. This will be analysed by the detection algorithm as a live stream (or post flight) and return the data for any detected objects, its predicted probability and location. These objects can then be removed from the runway manually or autonomously. A graphical representation of the concept of operations can be seen in Figure 1.

1. LITERATURE REVIEW

The detection of foreign objects for the current application in airports has been completed previously [10]. This has been used as the starting point and the basis to develop improvements. The paper used RGB images of the selected object classes and investigated the different deep learning models and the accuracy of the object detection. Using 1500 images of 5 different classes, the models were trained using Yolov3 and Microsoft Azure Custom Vision.

The results from the full-scale testing and base experiment found that the Microsoft model was significantly more accurate with a mAP of 82.9% compared to 28.2% for the Yolov3 model. The maximum height of the drone images was also investigated and found that a maximum height of 12m was possible with an accuracy of 91% for the Azure model compared to 3m with Yolov3.

The use of drones and object detection has also been used in other complex applications. In research study [11] a model was developed to detect bottles and their orientation with UAVs in inaccessible and remote locations. The relevant components of the paper explored the collection of images and the statistical analysis of the data used for training the selected models. The data collection followed four different suggestions of object variations within images. These are:

- Use a wide range of scale and aspect ratios
- Different background scenes
- Different orientations
- A wide range of object types

To vary the scale and aspect ratio of objects, the drone flew at a range of altitudes between 10 and 30m to collect the images. The paper selected eight different backgrounds /locations and ensured that the quantity of images in the database for each scenario is similar. This avoids overfitting the model or preferencing a specific location.

It can be seen in [12] that the paper used image detection from low altitude UAV images to locate rubbish. The paper discusses the success that algorithms can have when determining between distractions with similar objects, in this case rocks to litter. For the created model, it was able to reliably distinguish between rocks and rubbish. The data set was made up of 772 images with 3716 annotations. A large quantity of distractors were also included within each image, these are natural elements like stones and leaves.

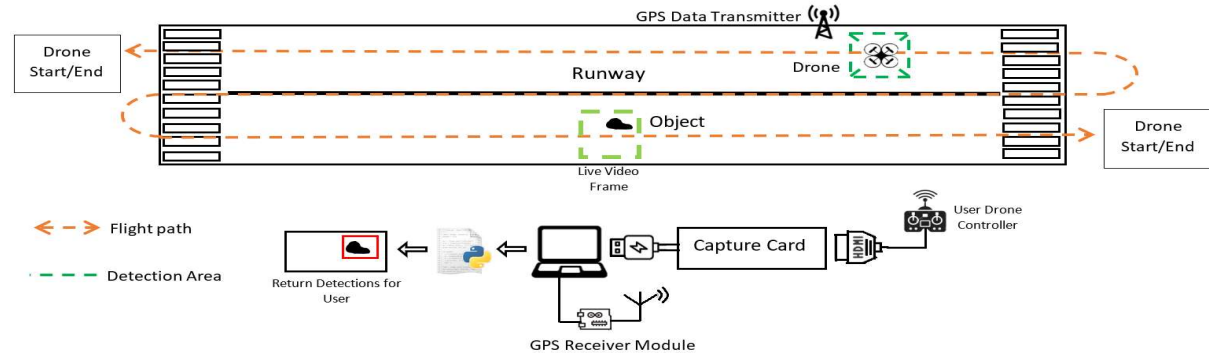


Figure 1: High Level Graphical concept of operation for the project.

Only a single ‘Rubbish’ class was used instead of selecting separate types of possible rubbish.

From the literature review it was evident that the quantity of images is only a single variable when considering improvements and accuracy of the detection model. Other considerations include the variation of objects, orientation, backgrounds, illumination and aspect ratio.

2. MATERIALS AND METHODS

The methodology involved three stages, these are summarised as follows:

- *Material and Data Acquisition*
The acquisition of materials and equipment required. The collection of the training images/data and prepared for training.
- *Model Training and Refinement*
During model training and testing, continuous refinement is required to ensure a balanced and accurate model.
- *Model Implementation (Testing and Live Detection with GPS)*
Model testing and analysis of results which concentrates on the implementation and testing with the Live detection and GPS location module

Materials and Hardware

A broad sample of FOD materials were selected for the project, including the objects shown in Figure 2. These were categorised based on the similarity in shape or material properties.

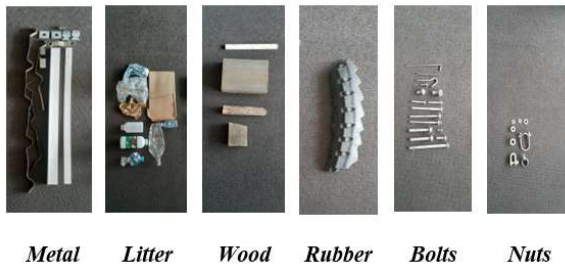


Figure 2: Example of selected objects.

While some objects may not be particularly destructive if ingested like a sheet of paper they can still be found on runways and need to be detected for removal. Other objects include large and small pieces of metal and bolts, which are one of the more common FOD found and can cause major damage if ingested by an engine.

The Mavic Air 1 (DJI, Shenzhen, China) was selected to capture the training images and complete the model testing due to the availability, manoeuvrability and cost as well as the relatively high specification of video imaging and gimbal components. The Phantom 4 Pro (DJI, Shenzhen, China) was acquired for the Live detection component testing. The Mavic Air measures 168×184×64 mm

unfolded and weights 430g compared to the Phantom 4 at 1388g and 350 mm diagonally across.



3a) DJI Mavic Air 1

3b) DJI Phantom 4 Pro

Figure 3: Selected drones.

Both drones can perform collision avoidance and the gimbal mechanism has a large range of movements. The camera properties are summarised in Table 1.

Table 1: Drone Camera Properties.

	FOV	Image Size	Video Resolution
Mavic Air	85°	4:3: 4056×3040 16:9: 4056×2280	4K: 3840×2160 2.7K: 2720×1530 FHD: 1920×1080 HD: 1280×720
Phantom 4 Pro	84°	3:2: 5472 × 3648 4:3: 4864 × 3648 16:9: 5472 × 3078	4K:3840×2160 2.7K:2720×1530 FHD:1920×1080 HD:1280×720

The video recording has different frame rate options for the different resolution options. For the Mavic Air 1, these range between 24-30 for the 4K while the 2.7K has the option to record up to 60 frames per second. The FHD and HD can be between 24 to 120 frames per second. [13] The Phantom 4 has a larger range of options with an increase in frame rates.

The other hardware components are used in the live detection component. To capture the live stream from the drone, a HDMI Video capture card was used. This captures footage from an external source camera and converts/encodes the video stream into a format that can be interpreted by the computer. [14]. The selected component was the Elgato Camlink 4k. This device supports a large number of video resolutions from 720 × 480 at 60 frames per second, up to 3840 × 2160 at 30 frames per second. It outputs the same resolution with no reduction in quality. [15]. Due to the larger output format, a USB 3.0 port with appropriate cables/adaptors were required. It also required a high-performance graphics card and processor in the computer.

The U-blox NEO-7M GPS module was selected. It is a high performance, small and light weight module with an approximate accuracy of 2.5m. The device requires a supply voltage of 3.3V. [16] The small and light weight Arduino Nano was selected to control, read and transmit the data from the GPS. To communicate between the Arduinos and the computer, a light weight, low power radio communication module was required. For testing and the live detection demonstration, the APC220 module was

selected. This has a high sensitivity and a functional range over 1000m within line of sight and requires a 5V voltage source. [17] The modules were configured and set up at a frequency of 434MHz. Both the radio and GPS were operated at a serial baud rate of 9600. The implementation of the GPS and radio modules is described in more detail in section 7. These primary hardware components are illustrated in Figure 4.

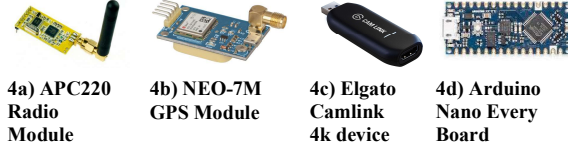


Figure 4: Main hardware components used.

3. RESOLUTION AND ACCURACY

The implementation requires an accurate representation and analysis of results and data. Specific components that relate to the resulting accuracy and resolution include the drone images resolution from the camera, the altitude, GPS coordinates data and the object positioning measurements within an image. The proposed effects of the weather conditions are also discussed.

Image/Object Resolution

The object resolution within an image can be found using the image resolution with the camera properties and the height of the drone. Using a specific object and the known drone height, the resolution of the object can be found and therefore the relationship between the pixel length and the height.

A bright blue test object was used which was 12cm in length and an image was taken at a height of 4m directly above the object. The resulting object in the image was approximately 87 pixels long. Therefore, the ratio is 7.3 pixel/cm. Taking additional images at increasing heights from 2-15m, the relationship Equation 1 was determined.

$$f(h_x) = 33.42 \times h_x^{-0.7687} - 4.519 \quad (1)$$

The resultant function adjusted R-squared value is 0.9983 and the RMSE is 0.3618. The relationship does not consider the object colour, background or other external circumstances of the environment that may affect the ability to 'see' the objects.

For comparison, a Phantom 4 Pro camera was also tested. Figure 5 shows the comparison between the drone height vs the object resolution. As expected, the resolution and pixel ratio decrease as the drone increases in height.

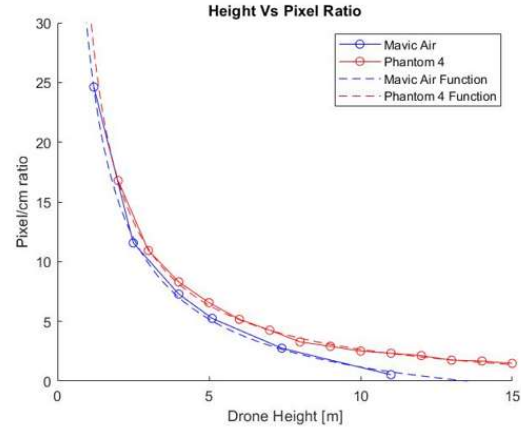


Figure 5: Drone height vs Object resolution.

The Brisbane International airport 01/19 runway has a length of 3560m and a width of 60m. These dimensions do not take into account the additional safety areas including a minimum 90m on either end and a 7.5m wide shoulder. [18] Using these dimensions, the required minimum height of the drone to capture the complete width of the runway can be found. The Mavic Air 1 camera has a Field of View (FOV) of 85 degrees. As a comparison the Phantom 4 Pro has a FOV of 84 degrees and a better camera resolution. The resulting application heights can be seen in Table 2.

Table 2: Physical application resolution and height results.

	Mavic Air 1	Phantom 4 Pro
Min height @ Runway Width [60m]	32.7m	33.3m
Min height @ Runway Width + Safety [75m]	40.9m	41.6m
Max height @ 1 Pixel/cm Resolution	10.4m	17.5m
Ground Width Covered @ Max height	19m	31.5m
Minimum Passes Required	4	3

Theoretical application

With the human eye image classification, it has been found that a minimum of 17×17 pixels representation is required to determine a face within an image and up to 47×47 pixels for complex images/objects and scenes. More research is required for detection of objects and computer vision applications. [19] Using the object resolution relationship and assuming that the absolute minimum resolution is 1 pixel per centimetre for detection, the resulting application requirements can be determined.

For the Mavic Air, a minimum height of 40.9m would be required to capture the extra safety boundary in the image. Assuming a minimum ratio of 1 pixel/cm, then the Mavic Air 1 can fly at a maximum height of 10.4m, the ground width of the image captured is 19m. Therefore 4

passes would be required to capture images of the complete runway and safety edges.

The improved camera on the Phantom 4 Pro allows for a maximum height of 17.5m and a field of view width of approximately 31.5m. Therefore, only 3 passes would be required with a small overlap between passes.

The proposed minimum time required for the Mavic Air drone can be approximated by assuming a constant maximum ground speed of 25km/h (6.94m/s), without effects of wind. The time required is 9 minutes per lap, including the two additional 100m safety areas. Considering the battery life of 21 minute, this would allow for 2 passes before requiring charging. Therefore, it would not be possible to complete the required 4 passes with a single drone on one battery charge. It is proposed that multiple drones would be required, a minimum of two to complete the passes simultaneously or to complete 2 passes at a time and charge between. Another possibility is to attach a larger external battery, or to fly faster. The Sports mode can travel at 68.4 km/h and would complete a pass within 3 minutes and therefore the 4 passes within 12min however the battery life at this speed is unknown and would need exploring.

GPS Accuracy

The GPS coordinates of the drone will represent the centre of the image. There are two formats for expressing the Latitude and Longitude position, these are Degrees, Minutes, Seconds (DMS) or Decimal Degrees (DD). The accuracy of the device is affected by the strength, quantity and quality of satellite signals found and other interferences. These include atmosphere conditions, buildings or obstructions. [20] A larger number (minimum of 4) and improved stability of the satellite signals can significantly improve the accuracy.

The representation accuracy of the GPS coordinates value depends on the number of decimals provided. Each additional decimal increases the accuracy by a magnitude of one hundred. This can be seen in the Table 3 below. [21]

Table 3: GPS decimal place accuracy.

<i>Decimals</i>	<i>Degrees</i>	<i>Distance</i>
0	1.0	111km
1	0.1	11.1km
2	0.01	1.11km
3	0.001	111m
4	0.0001	11.1m
5	0.00001	1.11m
6	0.000001	0.111m

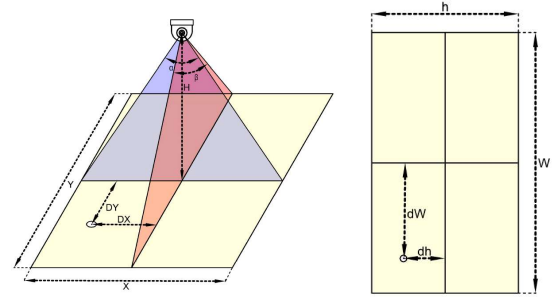
A value presented to 5 decimal places will result in a position within an area of 1.11m. See the Live Detection Process and Results Section 8 for the static GPS testing results.

Image Object Positioning

The coordinate in real space of an object bounding box centre can be found with Equation 2. The coordinates (DX , DY) in meters relative to the centre of the image using the distance of the object in pixels from the centre of the image (dx , dw). The accuracy of these location measurements are dependent on the height (H) and the accuracy of the altitude measurement. [12].

$$DX = 2H \frac{dw}{w} \tan\left(\frac{\alpha}{2}\right), \quad DY = 2H \frac{dh}{h} \tan\left(\frac{\beta}{2}\right) \quad (2)$$

See Figure 6 for a graphical representation of the variables. The resulting y-coordinate will be reversed as the image is flipped when viewing the pixels within the image



6a) Observed World Coordinates 6b) Captured Image coordinates

Figure 6: Calculate the world coordinates of the object bounding box centre based from centre of image.

Weather Affects

The local weather conditions are a significant constraint on the implementation of the object detection process and the proposed implementation with a drone. Different weather conditions will affect the detection model as well as the drone flight and camera performance. The Mavic Air 1 has a documented max wind speed resistance of 29 – 38km/h, however the performance and controllability will be significantly reduced. [13] The impact of rain and mist, or moisture in the air can also be damaging to the longevity of the drone performance due to the exposure of the electrical components, and the drone manufactures highly recommend against flying in these conditions.

The object detection results can be significantly affected by the weather conditions including the changes in the sun position and intensity during the flight compared to the training images used. These include changes to the sunlight if early/late in the day or a cloudy or foggy conditions. The effects of rainwater on the ground and camera surface may also provide false/missing detections.

There are many research papers on improving object detection models with various challenging weather conditions, especially in the field of autonomous vehicles. In [22] they used a Visibility Complementation Model to analyse the image and assess the weather 'state' and apply

a correction before the detection model analysis. This found that this approach is successful without the need to retrain the model. As seen in [23], [24] also developed a method to remove the effect of weather in images and the detection process with an improvement in the detection success.

These weather affects can be further investigated and may be overcome with additional training, image processing and drone design modification.

4. DATA COLLECTION LOCATIONS

The collection of the data for the training and testing of the model was conducted on multiple occasions at three different locations. The locations satisfied the local and federal government regulations to permit drone flights. All QUT safety regulations and risk assessments were also completed. A flight plan and risk assessments were completed for each location and a flight logbook and checklist has been maintained.

For the data collection, the objects were randomly distributed and arranged within the designated survey area selected. The weather conditions and time of day for each flight varied, however were all conducted with minimal wind, very good visibility and no rain.

Site 1: Private Property, Noosa QLD

This location is a private property with landowner approval to fly the drone, see Figure 7. The property has an abandoned private airstrip runway that makes it ideal and realistic to capture the images. The runway is 550m long however access was only was granted for the first 100m.



Figure 7: Private Property Drone Survey Area.

Site 2: Drone Approved Park, Brisbane City, QLD

The second location was a Brisbane City Council drone approved park, see Figure 8. There is a large flat (unmarked) carpark. The area available is approximately $62 \times 36\text{m}$.



Figure 8: BCC Location Drone Survey Area.

Site 3: Drag Strip, Ipswich, QLD

A third location was used for the Live testing. To fly a drone at this location, flight approval was required from the business management, and due to the location, approval was also required from the local RAAF base, 8km away. These permits were received well before flight day. The drag strip is 1km long and 16m wide and is an ideal representation for a runway. See Figure 9 for the survey area.



Figure 9: Drag Strip Location Drone Survey Area.

5. DATA COLLECTION AND ANALYSIS

The FOD test objects used were classified into 6 main types include Litter, Wood, Bolt, Metal, Rubber and Nuts. These were selected to represent the most likely objects that may be found on a runway and within an airport. The specific objects for each class vary in size, orientation, colour, shape and texture. This increased the complexity and contributed to the image variation and diversity. See Figure 2 above for an example of the FODs used.

The drone was flown at a range of heights between 2m and 15m to capture the individual images. The objects were rearranged and moved randomly to ensure a large and diverse data set of training images.

These images were uploaded into the Microsoft Azure Custom Vision database and labelled with the corresponding FOD class tag. An example can be seen in Figure 10 where the five objects have been manually identified and tagged.



Figure 10: Tagging objects example.

Throughout the collection and training stages, an analysis on the quantity of objects in each class was compared to ensure a balanced quantity was used to minimise object/training bias. The database was trained with the selected properties for the desired model, as described below.

Detection Models

Two different detection models have been created. The first model created has 6 different classes for each of the major FODs used. A second model created has only a single object class called "Object". This is realistic in implementation as the exact material, properties and size of the detected object are less important, and all FODs will want to be removed from the runway when detected.

To use the model to detect objects in an automated way the python script is required to access the Azure custom vision model. This can be completed in two ways. The first is to publish the model online, which creates an API prediction key, id and URL that can be called and used in python to upload images and return the results. This uses the model online, and throughout the process it uploads the image to the online database and returns the detection data. This process is slow and requires a reliable high-speed internet connection. The image results are saved to the database and can be analysed at a later time or used to further train the model. The image size is restricted to 4MB so large images or resolution types cannot be analysed without being compressed or altered first.

The second method is to export a local copy of the model, this required training the model in the compact format. However, this significantly reduces the overall accuracy of the model. The compact model is exported as a TensorFlow model and contains a file of the class labels and the model.

The multiple class model database has over 200 images for each class and the single class model has over 800 images.

The resulting training performance characteristics can be seen in Table 4 for the two different training process and the two detection models. A critical parameter is the Probability Threshold, this sets the minimum percentage the model will accept the detection result and to classify the detection as successful or ignore the results. The performances characteristics are based on these successful points and if they were correctly identified. The probability threshold is set to 70% for the model comparison below.

Table 4: Compare the training process and detection models performance.

	Precision [%]		Recall [%]		mAP [%]	
	General	Compact	General	Compact	General	Compact
Multi	77.0	96.3	72.7	5.6	77.5	40.7
Single	85.7	88.8	90.5	37.2	94.0	55.4

The precision value is the probability a predicted tag is correct. The recall value is the percentage of tags the model correctly found and the mean average precision (mAP) is the overall performance and accuracy of the model.

When comparing the different model performance characteristics, the single class model has better results for all three properties. The online full model has a very high recall value and mean average precision, the single class compact results are significantly better than the original compact model.

These model results are accurate and robust for detecting all types of objects especially when there are no natural distractions. When testing with additional distractions (leaves) included in the image, the detection accuracy decreases with an increase in false positive results. Through additional training iterations this was minimised

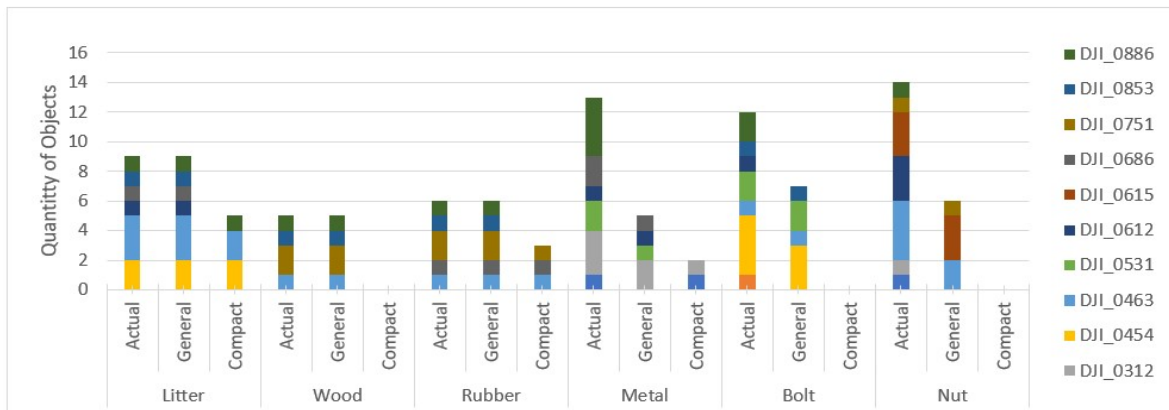


Figure 11: Compare the correct object detection results with the multi class model.

and with more training the model could be further improved.

6. MODEL TESTING

Multiple model iterations were created and tested. The model bias or imbalances found were improved for the next iteration. The current models were tested using individual images with an altitude range between 2.9m and 11.7m, the probability threshold was set to 50%, all images were compressed before testing the general online models. The images identified as DJI_0244 and DJI_0271 in the tables below were unable to be tested as the file size was very large and exceeded the maximum size even after using an image compressing software.

Figure 11 visually displays the actual object quantity within each image and the correctly returned results for each object class for the multi class general and compact training models. The litter, wood and rubber classes were all found with the general model while no nut, bolt and wood objects are correctly detected using the compact model.

For the single class model the actual and general results are the same for many of the testing images with only 2 images with errors, see Figure 12. The compact object count is significantly lower for all images, with 4 test images with no objects detected above the threshold of 50%. These 4 images are at a range of heights and contain many of the smaller objects including nuts and bolts.

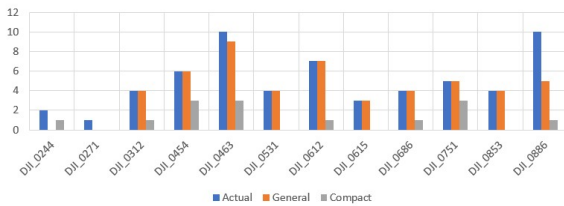


Figure 12: Compare the correct object detection results with the single class model.

It can be seen that for both models, the general online training is significantly more accurate at detecting the correct object class, as expected from the performance characteristics. The error rate for the compact models is above 80% due to the missed objects compared to the 32.1% for the general multi 6-class model. The objects that are detected with the compact model are largely correct as expected due to the high precision value. The single class general trained model had a low error rate of 10.5% with 51 correct objects detected out of 57.

An example image and model detections can be seen below. Figure 13 is the image identified as *DJI_0454* and Figure 14 is *DJI_0751*. Both examples are at different heights and with a variety of objects. Both the general models have correctly identified all the FODs in the image, and the compact model did not identify the other bolt objects.

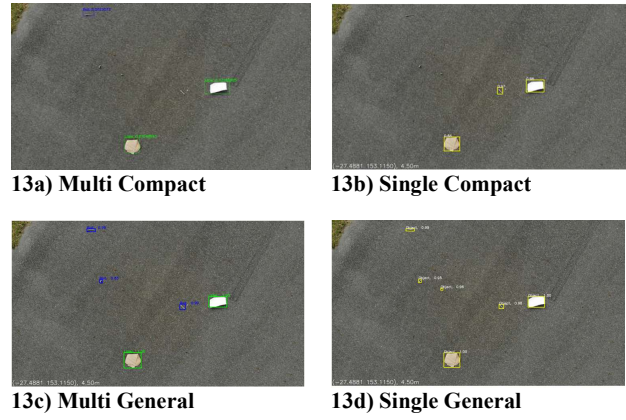


Figure 13: Example detection results, DJI_0454 image at 4.5m altitude.

The two general models have correctly identified all the FODs in the image and the compact models have both missed the small objects.

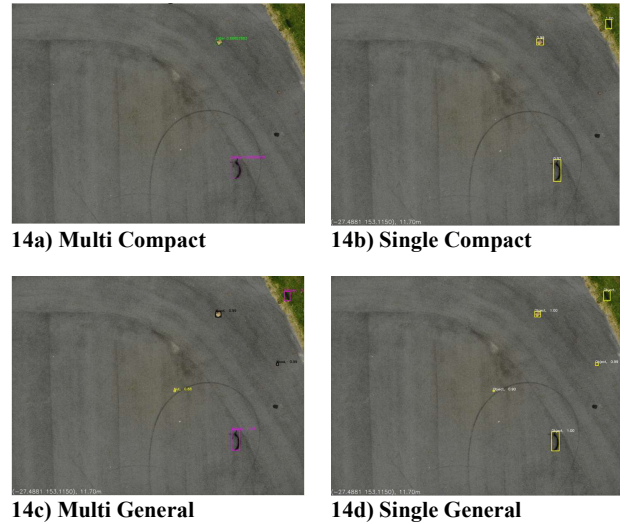


Figure 14: Example detection results, DJI_0751 image at 11.7m altitude.

For all testing results, the drone height was a significant factor with the accuracy of the correct detections, but this was also depended on the size of the object. The models can be further improved with additional training and data collection.

A video demonstration of the different detection models was created to compare the accuracy and consistency of the results. The drone was flown at a height of 10m on a clear day. The video can be found with this link: <https://www.youtube.com/watch?v=0aCmPBZITss>.

The GPS data saved in the image meta data is to 4 decimal places with an approximate horizontal accuracy of 11.1m. This was considered very inaccurate and further improvements were made for the live testing and detection component.

The location of the objects can be found with respect to the centre of the image as described previously. See Figure 15

for an annotated example two object detections with the calculated position dimensions. The accuracy is based on the height data accuracy and will be more subject to error at lower altitudes.

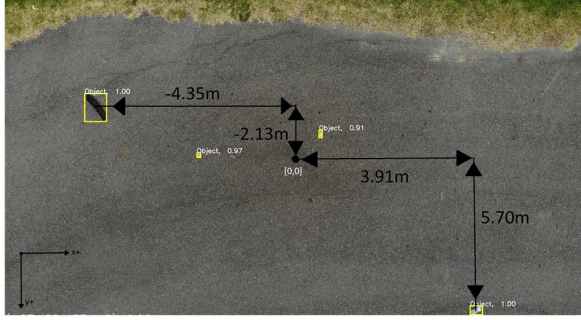


Figure 15: Example Object Position Results.

7. LIVE DETECTION PROCESS

A vital component of the removal of FODs is the minimal time available to detect and collect FODs due to the large quantity of flights and very short transition times between flights. This applies a time constraint on the detection process to be fast and efficient. To decrease the processing time, a live detection process was created. Through multiple iterations the preferred method was to use the Elgato Camlink 4k device to capture the video stream from the remote controller while maintaining the stream quality with very minimal video stream latency. This process only requires the capture card and does not require any additional processing or software on the computer. A specific HDMI adaptor is required on the drone controller, which is only available for specific professional drone models.

The single class with the compact training model was selected due to the relatively high accuracy compared to the multi-class model and the fast analysis time. The process would remain the same for any trained compact model, with minor code changes required when using the general online model.

The detection process is the same as before except the individual frames are provided by the camlink, these are analysed by the detection model with the probability and pixel location data returned. For images with detection results larger than the set probability threshold, the image frame is saved with the corresponding bounding box, label and probability annotations included. The delay caused by the analysis is approximately 3 seconds with the compact local model and 10 seconds for the online general model. These times depend on the resolution and size of the image/video stream. The Camlink provides the current frame from the input device so all frames between detections are lost. The ground speed of the drone will be especially relevant to reduce the chance of an object being completely missed. The goal was to provide a minimum of three image opportunities within a set distance so objects

can be detected. This is based on the drone height to set the maximum ground speed.

The camera image footprint can be found with the specific camera properties is Equation 3 and can be used to find the relationship with the drone height. [25]

$$X = 2H * \frac{\tan\left(\frac{\alpha}{2}\right)}{\sqrt{1+r^2}}$$

$$Y = 2 * r * H * \frac{\tan\left(\frac{\beta}{2}\right)}{\sqrt{1+r^2}}$$
(3)

Where:

- r is the image size ratio
- H is the drone height [m]
- α, β are the camera FOV angles
- X, Y are the physical dimensions [m]

For example, at an altitude of 8m, the resultant ground coverage is $12.8m \times 7.2m$ with the 16/9 image ratio and the FOV of 85° . To allow for a three second model delay and three detection frames (approx. 10 second flight time), then the maximum ground speed is 0.72m/s. A slower speed will increase the quantity of frames analysed per meter and a faster speed may reduce the opportunities the model will have to detect the objects.

To further improve the accountability and usability, a GPS module was added to record the location of the drone. The selected NEO-7M module collects the satellite data and calculates the coordinates and other properties. These GPS values are streamed to the radio serial port. The radio module sends the serial information to the second Arduino Nano to read. The python script then reads the open serial com-port to extract and decode the GPS data. The communication between the Arduino devices uses the Software Serial library. This allows for multiple devices to be connected by using the digital pins.

To power the microcontroller on the drone, a 9V battery was wired up with a SPST switch and LED to indicate power. The Arduino can be powered by a regulated 5V or 6-20V unregulated external supply [26]. The GPS and radio are connected to the required Arduino regulated output voltage pins and a common ground.

The resulting transmitter and GPS module circuit and wiring diagram is shown in Figure 16.

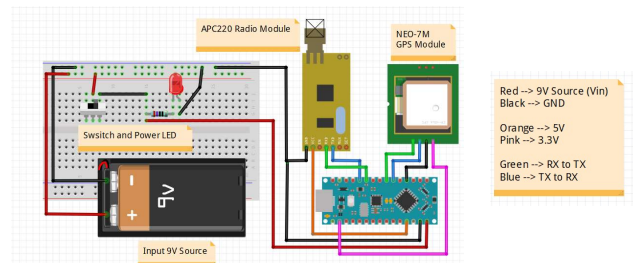


Figure 16: Transmitter and GPS Circuit.

The receiver circuit is connected and powered by the 5V USB port from the computer. The wiring diagram is shown in Figure 17.

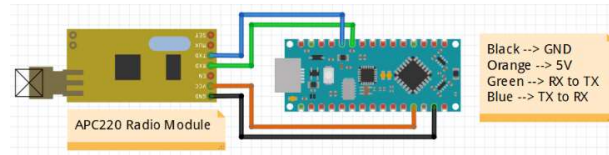


Figure 17: Receiver module Circuit.

The receiver module was mounted in a light-weight polycarbonate enclosure to help contain and protect the components. It also minimises any possible interference or problems while in flight. This can be seen in Figure 18 with the top cover removed.

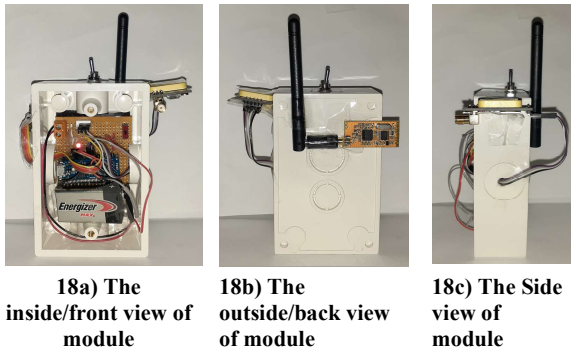


Figure 18: Final constructed receiver module and storage.

The resulting GPS data is saved in multiple file formats to be user friendly and readily accessible. An example is the Google Earth 'KML' file in Figure 19 where the GPS coordinates of the drone are saved for the detected frames, a pin is placed as well as a summary of the quantity of objects found and the image identification name. Both 'CSV' and HTML files are also created with the object detection information. The HTML text file is easy to use and only requires a small amount of internet access compared to loading Google Earth. However, the map cannot be zoomed in to the same extent and limits the position perspective. The edited video frame with the printed bounding box and detection information is also created that can be manually viewed as with the previous process.



Figure 19: Example Google Earth GPS data results.

8. LIVE DETECTION RESULTS

To test the single class detection model and the live detection process the DJI Phantom 4 Pro drone was used, because the original Mavic Air 1 drone does not include the required HDMI adaptor in the controller to connect to the video feed. This change in drone and camera had no significant effect on the accuracy of the model detection results.

The Arduino transmitter model was attached to the drone base. In total, with the 9V battery it weighed 200 grams. The drone had no issues with flight control with the additional weight.

A neutral location was selected as an effective test for the versatility and accuracy of the model without any bias towards a specific area or object. It is expected that the model would perform better when tested against similar environments to the training. The third location drag strip was selected for the testing. The drone was flown at a constant specified height with the live video stream, GPS coordinates and object detection results collated.

Static GPS Testing

Before the implementation of the module, a static GPS test was conducted. The resulting coordinate positions and quantity of satellites can be seen in Figure 20. Only the trailing 3 digits are presented. This demonstrates the importance of a large number and consistency of satellites detected. The position coordinates returned the largest changes when the satellite number dropped. Changes in the fourth decimal place approximately represents 11.1m in the accuracy of the returned position. The test was conducted over a 12-minute period.

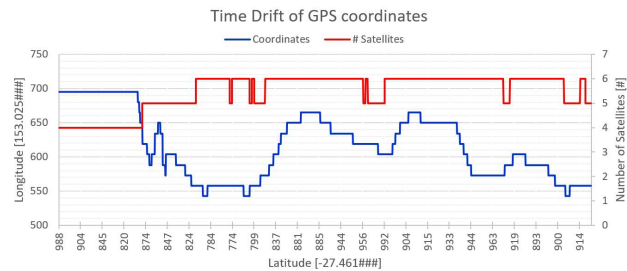


Figure 20: Static GPS test results.

Live Detection Results

The flight day weather conditions were sunny with very minimal wind and cloud cover. The flight tests were conducted at constant heights with the same object positioning. The maximum drone ground speeds were set at 0.54m/s and 1.1m/s to allow for the maximum model delay and the minimum of 3 frames for detection.

The example results 20a, 20b are taken on the drag strip and 20c, 20d are in the start area. The images are of the same objects in similar orientations. The 20b video frame successfully detected the two objects while the lower flight

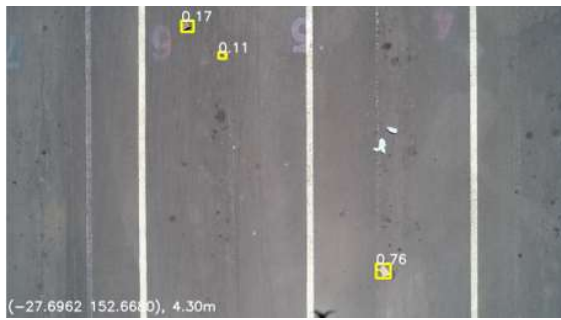
20a falsely detected an FOD and found the correct object with low probability. For the start area frames, these were similar to the training images and returned more objects with a higher percentage, however both heights missed objects.



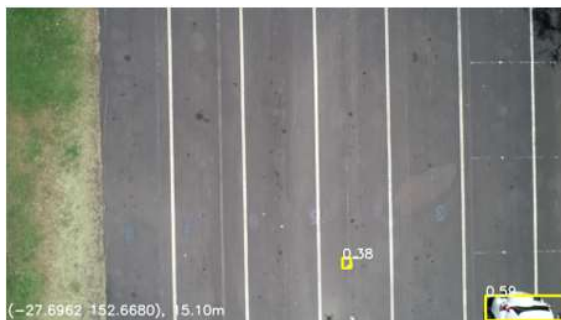
21a) Low Flight 1



21b) High Flight 1



21c) Low Flight 2



21d) High Flight 2

Figure 21: Live detection process; example video frame results.

The overall accuracy is entirely related to the accuracy of the trained model used; this is as expected from the previous testing results. The use of the GPS transmitter and receiver module significantly increased the GPS accuracy and allowed for real time updates of the image video feed.

A returned Google Earth map file with the detected objects can be seen in Figure 22.



Figure 22: Resulting Google Earth Map of objects detected and locations.

This testing proved that a live detection process can be completed to detect objects with minimal time processing delay and a user-friendly accurate coordinate data can be created and displayed.

Live Testing Detection Challenges

The testing at the Drag Strip provided a few challenges for processing and analysis. These challenges are a realistic representation of problems that may be present at an airport runway.

Due to the significantly amount of burnt rubber on the track it became very reflective when capturing the videos and a shadow of the drone was also present in a majority of the frames. These conditions can be seen in Figure 23.



Figure 23: Example environmental interference in images.

The shadow was detected in many images due to the increase in drone size with the Phantom 4 Pro. No objects were able to be detected when near or inside the reflective area of an image. These conditions can be further explored to avoid false positive detections in the future with additional training or changing the camera angle.

Due to the consistency of the rubber on the track many instances of small scratches or marks were falsely detected as objects.

9. CONCLUSION

The paper has covered the implementation of using a drone to detect FODs at an airport runway. The background literature review of significant components has been

conducted including the review of a previous report that provided inspiration for this paper. The required materials and hardware for the project was described as well as the resolution and accuracy concepts. The data analysis identified the relationship between the drone height and the object pixel resolution, and this was used to find the required height to successfully view the complete width of the Brisbane runway. Including the safety width, an altitude of 40.9m would be required for the Mavic Air 1. A minimum of 4 passes would be required to completely view the runway with the assumptions that a minimum of 1 pixel per centimetre is acceptable. The standard drone battery would not provide enough power to complete the detection in a single charge flying at max speed of 25km/h.

The three test sites were approved for drone flights with data collection over multiple dates. These have been uploaded to the Microsoft Azure Custom Vision database and tagged appropriately. Two different models have been created with two different training methods. The online general model has a maximum file size limit of 4MB and requires images to be compressed before training. The compact model is significantly faster for the analysis, however it loses accuracy in the resulting detections. The multi-class model has over 200 images for each class and the single model has over 800 images.

The testing of the models found the general models to be accurate and detecting a significant number of the objects. The single general model had an error rate of 10.5% and 32.1% for the multi class model with the testing conducted. The compact single model has an error rate of 80%.

The actual position locations dimensions have been determined based on the altitude of the drone with the GPS image meta data extracted. This has an accuracy of 11.1m and was improved with the implementation of the GPS transmitter module to 2.5m. This was applied for the live detection process. This paper explored the implementation of the live process and the drone speed and altitude choices to minimise missing frames due to the delay in the analysis. The testing results found that some objects were correctly detected with an increase in false detections due to the neutral location. The GPS coordinates were used to create a user-friendly accessible format to find the detected objects and the results for collection.

This paper proved the viability of a live detection process and improved on the previous research accuracy results.

There are many possible developments for the project in the future, which includes:

- Improved accuracy of the model and location coordinates;
- Multispectral camera for detection;
- Increase model and object complexity, natural objects;
- Include / test additional weather scenarios;
- Data augmentation in training (image modification, Gaussian blurring, darkening, lightening or mirroring);
- Reduce the delay time between detections, e.g pre-image processing and/or threading;

- On board processing of video stream and object detection.

A combination of these developments may further increase the projects accuracy and implementation concept.

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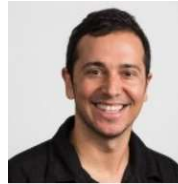
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Peter Trotter holds a BEng (Chem) and has a strong interest in aviation including building and flying light aircraft, sailplane racing and UAV commercial operations focusing on live streaming object identification.